

PROJECT_REPORT.md

Separate PDF version of **PROJECT_REPORT.md** for the Daily Habit Monitoring System project.

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# PROJECT REPORT
# Daily Habit Monitoring System
```

1. Introduction

The Daily Habit Monitoring System is a desktop-based application developed using Python, Tkinter and MySQL. The purpose of the project is

2. Objectives

1. Provide user registration and login.
2. Allow users to create personal habits.
3. Store habits in a MySQL database.
4. Search, update and delete habits.
5. Record daily completion status.
6. Display stored records through a graphical interface.
7. Demonstrate Python, Tkinter and MySQL concepts.

3. Technologies Used

Python

Used for application logic, functions, conditions, loops, exception handling and database operations.

Tkinter

Used to create the desktop graphical user interface.

MySQL

Used for persistent storage of users, habits and daily records.

mysql-connector-python

Used to connect Python with MySQL.

4. System Architecture

The application has three main layers:

1. GUI layer – Tkinter screens and widgets.
2. Application layer – Python functions and validation.
3. Database layer – MySQL tables and SQL queries.

5. Database Design

users

Field	Type	Description
user_id	INT	Primary key
username	VARCHAR(50)	Unique username
password	VARCHAR(100)	User password

habits

Field	Type	Description
habit_id	INT	Primary key
user_id	INT	Foreign key
habit_name	VARCHAR(100)	Habit name
category	VARCHAR(50)	Habit category
target	VARCHAR(100)	Daily target

habit_records

Field	Type	Description
record_id	INT	Primary key
habit_id	INT	Foreign key
record_date	DATE	Date of record
status	VARCHAR(20)	Completed/Not Completed

6. Relationships

One user can have many habits.

One habit can have many daily records.

The relationships are implemented using foreign keys.

7. Python Concepts

The project demonstrates:

- Variables and data types
- Conditions
- Loops
- Functions

- List
- Tuple
- Set example
- Dictionary
- File handling example
- Exception handling

8. Tkinter Concepts

The project demonstrates:

- Login screen
- Registration/form
- Labels and Entry
- Buttons
- Combobox
- Radio buttons
- Treeview
- Search
- Update/Delete
- Multiple screens
- Message boxes

9. MySQL Concepts

The project demonstrates:

- Database creation
- Table creation
- INSERT
- SELECT
- UPDATE
- DELETE
- WHERE
- ORDER BY
- GROUP BY example
- Primary key
- Foreign key
- JOIN
- LIKE/search filtering

10. Functional Modules

Registration

Creates a new user account.

Login

Checks the entered username and password.

Add Habit

Stores a habit with category and target.

View Habits

Displays the user's habits.

Search Habit

Searches habit names using SQL LIKE.

Update Habit

Changes habit information.

Delete Habit

Deletes a selected habit.

Mark Completed

Stores or updates today's status.

Daily Records

Displays recorded dates and statuses using a JOIN.

Logout

Clears the current session and returns to login.

11. Expected Results

The system successfully provides a GUI-based habit monitoring workflow.

Example:

```

```text
Login
↓
Dashboard
↓
Add Habit
↓
View/Search/Update/Delete
↓
Mark Completed
↓
Daily Records

```

...

## ## 12. Advantages

- Easy to use
- Graphical interface
- Persistent database storage
- Search functionality
- CRUD operations
- Daily tracking
- Demonstrates multiple programming concepts

## ## 13. Limitations

- Passwords are stored as plain text in this academic version.
- No cloud synchronization.
- No password recovery.
- No charts or advanced analytics.
- Single local MySQL installation is assumed.

## ## 14. Future Enhancements

- Password hashing
- Habit streak calculation
- Progress charts
- Weekly/monthly reports
- Notifications
- Export to CSV/PDF
- Password reset
- Multiple themes
- Cloud database support

## ## 15. Conclusion

The Daily Habit Monitoring System demonstrates how Python, Tkinter and MySQL can be combined to create a functional desktop application.